

Experiment No.	2
Experiment Title	Email Spam/Ham Classification using Naïve Bayes and KNN ¹
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1 Objective

- Build spam classifiers using Naïve Bayes and K-Nearest Neighbors (KNN).
- Compare Gaussian, Multinomial, and Bernoulli Naïve Bayes variants.
- Study the effect of varying k on KNN performance.
- Compare the KDTree and BallTree neighbor-search algorithms.
- Optimize KNN using GridSearchCV and RandomizedSearchCV.
- Evaluate computational (training/prediction) time and theoretical complexity.
- Evaluate all models using 5-Fold Cross-Validation.

2 Problem Statement

Given a collection of 4601 emails represented by 57 numeric features derived from word frequencies, character frequencies, and capital-letter run-length statistics, the task is to build a **binary classifier** that labels each email as *spam* (1) or *ham* (0). Two families of classifiers are studied: probabilistic Naïve Bayes classifiers (Gaussian, Multinomial, Bernoulli) and the instance-based K-Nearest Neighbors algorithm. Beyond obtaining a well-performing classifier, the aim is to understand *why* each algorithm behaves the way it does on this dataset, covering accuracy, precision/recall trade-offs, hyperparameter sensitivity, and computational cost.

3 Dataset Description

Dataset Name	Spambase
Dataset Source	UCI Machine Learning Repository / Kaggle (https://www.kaggle.com/datasets/somesh24/spambase)
Number of Samples	4601
Number of Features	57 (48 word-frequency, 6 character-frequency, 3 capital-run-length features)
Number of Classes	2 (Spam = 1813, 39.40%; Ham = 2788, 60.60%)
Missing Values	0 (verified, no imputation required)
Train-Test Split	80% train (3680 samples) / 20% test (921 samples), stratified on the class label

¹GitHub Repository: <https://github.com/blackscythe123/Machine-Learning-course>

4 Exploratory Data Analysis

Exploratory Data Analysis Summary — Spambase Dataset

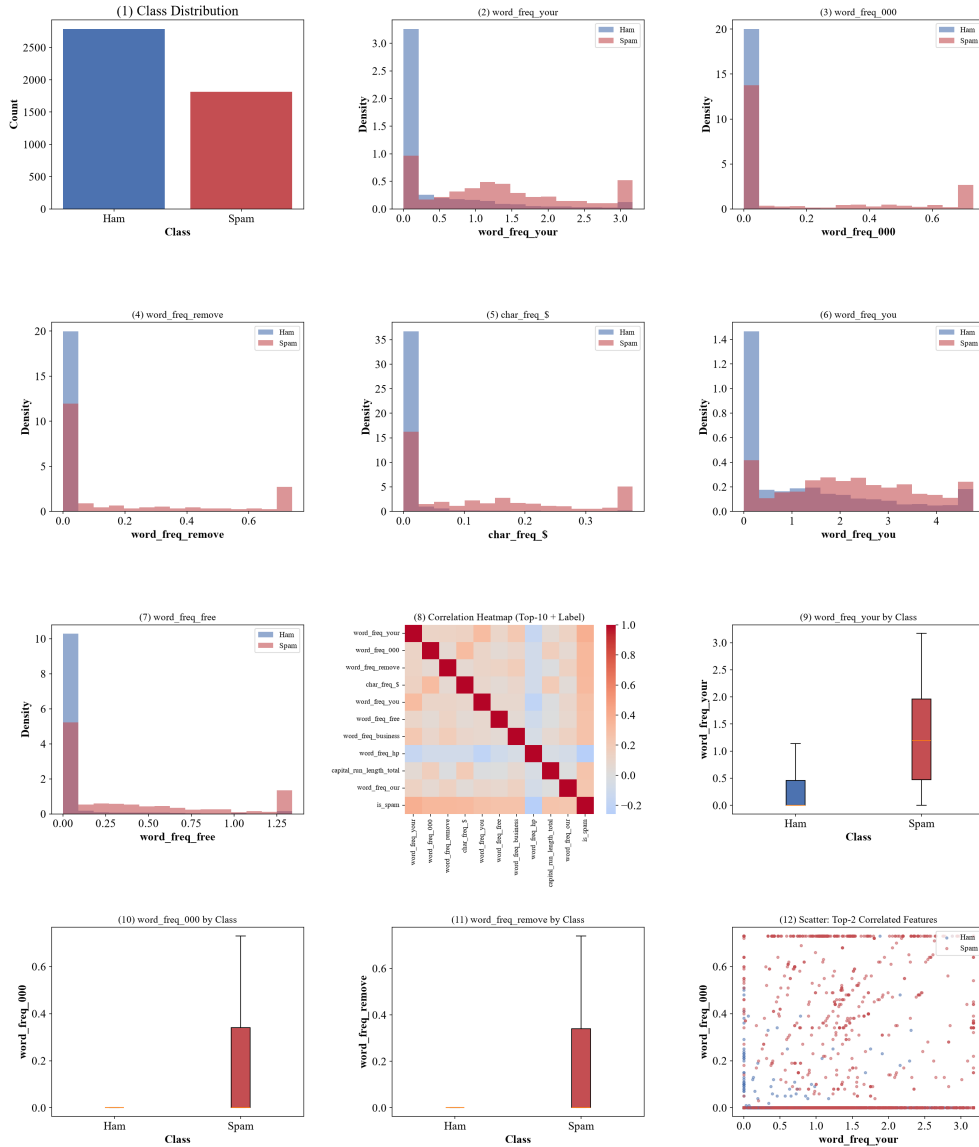


Figure 1: Consolidated exploratory data analysis (12 subplots, single page): (1) class distribution; (2–7) distributions of the six most class-correlated features, split by class and clipped at the 95th percentile; (8) correlation heatmap of the ten most class-correlated features; (9–11) boxplots of the three most discriminating features by class; (12) scatter of the two most correlated features, coloured by class. Exported at 600 DPI as `figures/00_eda_summary.eps` and reproduced here as PNG.

Inference (subplot 1, class distribution): The dataset contains 2788 ham emails (60.6%) and 1813 spam emails (39.4%), a moderate class imbalance rather than a severe one. Because the imbalance is mild, accuracy remains a reasonably informative metric, but precision, recall and F1-score are still reported throughout to avoid a misleading picture, particularly since a spam filter’s real-world cost of a false negative (spam reaching the inbox) differs from that of a false positive (a genuine email marked as spam). Stratified sampling was therefore used for every train-test split and cross-validation fold so that both classes are represented proportionally in each partition.

Inference (subplots 2–7, feature distributions): Spam emails show a visibly higher density at non-zero values of `word_freq_free`, `word_freq_your`, `char_freq_!` and `char_freq_$`, while ham emails are heavily concentrated near zero for these features. `capital_run_length_average` is also right-shifted for spam. All features are strongly right-skewed with a long tail of large values, which is the reason Gaussian Naïve Bayes (which assumes normally distributed features) is expected to under-perform relative to Bernoulli Naïve Bayes on this data.

Inference (subplot 8, correlation heatmap): Among all 57 features, `word_freq_your` ($r = 0.383$), `word_freq_you` ($r = 0.274$), `word_freq_free` ($r = 0.263$), and the two capital-run-length features ($r \approx 0.22$ – 0.25) show the strongest positive correlation with the spam label. This matches intuition: spam emails tend to address the reader directly (“you”, “your”), advertise free offers, and use long runs of capital letters for emphasis. No pair of features shows extreme multicollinearity ($|r| > 0.9$), so no features were dropped prior to modelling.

Inference (subplots 9–11, boxplots): The interquartile ranges for `word_freq_free`, `word_freq_your` and the character-frequency features are consistently higher and wider for the spam class, which makes them strong discriminating features. Ham emails have near-zero medians for almost every feature shown, which explains why simple thresholding on a handful of features (as Bernoulli NB effectively does after binarization) already achieves strong separation.

5 Data Preprocessing

- **Missing values:** The dataset was checked with `isnull().sum()` and confirmed to contain zero missing entries, so no imputation was necessary.
- **Encoding:** All 57 features are numeric (frequencies and run-length statistics), so no categorical encoding was required.
- **Feature scaling:** Two parallel versions of the feature matrix were prepared:
 - A **standardized** version (`StandardScaler`, zero mean / unit variance fitted on the training set only) used for **Gaussian NB** and **KNN**, since both algorithms are scale-sensitive.
 - The **original non-negative** version used for **Multinomial NB** and **Bernoulli NB**, since these variants require non-negative (count-like) input and standardization would introduce negative values that violate their assumptions.
- **Train-test split:** An 80/20 stratified split (`random_state=42`) was used, giving 3680 training and 921 test samples with class proportions preserved.

- **Feature engineering:** No additional feature engineering was performed; the 57 provided attributes were used as-is, consistent with the standard Spambase benchmark.

6 Machine Learning Algorithm

Naïve Bayes

Naïve Bayes is a probabilistic classifier based on Bayes' theorem with a conditional independence assumption between features given the class:

$$P(C | X) = \frac{P(X | C) P(C)}{P(X)}, \quad P(X | C) = \prod_{i=1}^d P(x_i | C)$$

Three variants differ in how $P(x_i | C)$ is modelled:

- **Gaussian NB** assumes $x_i | C \sim \mathcal{N}(\mu_C, \sigma_C^2)$, appropriate for continuous, roughly bell-shaped features.
- **Multinomial NB** models x_i as counts drawn from a multinomial distribution, naturally suited to word/token counts.
- **Bernoulli NB** binarizes each feature ($x_i > 0 \Rightarrow 1$, else 0) and models presence/absence of a term, explicitly penalizing the *non-occurrence* of a feature as well as rewarding its occurrence.

Advantages: Extremely fast to train (single pass, closed-form parameter estimates), works well with high-dimensional data, requires little training data. **Limitations:** The conditional-independence assumption rarely holds exactly; Gaussian NB is sensitive to non-normal feature distributions (as seen in this dataset).

K-Nearest Neighbors (KNN)

KNN is a lazy, instance-based learner. To classify a query point x , it finds the k training points closest to x (by a distance metric such as Euclidean or Manhattan distance) and assigns the majority class among them:

$$\hat{y} = \arg \max_c \sum_{i \in N_k(x)} \mathbb{1}(y_i = c)$$

where $N_k(x)$ is the set of k nearest neighbors, optionally weighted by inverse distance (`weights='distance'`). Neighbor search can be performed by **brute force** ($O(nd)$ per query), or accelerated with spatial index structures:

- **KDTree** – partitions the feature space along axis-aligned hyperplanes; efficient in low-to-moderate dimensions.
- **BallTree** – partitions data into nested hyperspheres; more robust in higher dimensions than KDTree.

Advantages: No training phase, naturally handles multi-modal class boundaries, simple to understand. **Limitations:** Prediction is expensive at scale, sensitive to feature scaling and irrelevant features, and both KDTree and BallTree degrade toward brute-force performance as dimensionality increases (the “curse of dimensionality”), a relevant concern here since $d = 57$.

7 Experimental Setup

Python Version	3.12.3
Scikit-Learn Version	1.8.0
IDE	Jupyter Notebook
Operating System	Windows 11
Hardware	Intel Core Ultra 9, NVIDIA RTX 5060

8 Hyperparameter Selection

Parameter	Search Space
KNN – n_neighbors (Grid)	{1, 3, 5, 7, 9, 11, 13, 15}
KNN – n_neighbors (Random)	{1, 2, ..., 25}
KNN – weights	{uniform, distance}
KNN – algorithm	{kd_tree, ball_tree}
KNN – metric	{euclidean, manhattan}
Cross-validation	5-fold stratified (StratifiedKFold)
RandomizedSearchCV iterations	25 (of 128 possible combinations)

Parameter	GridSearchCV	RandomizedSearchCV
Best k	9	8
Metric	manhattan	manhattan
Weights	distance	distance
Algorithm	kd_tree	ball_tree
Best CV Accuracy	0.9253	0.9255
Execution Time (s)	58.52	23.77

9 Results

Naïve Bayes Comparison (Test Set)

Metric	Gaussian	Multinomial	Bernoulli
Accuracy	0.8328	0.7763	0.8762
Precision	0.7146	0.7199	0.8716
Recall	0.9587	0.7080	0.8044
F1-Score	0.8188	0.7139	0.8367
ROC-AUC	0.9376	0.8248	0.9496

KNN: Accuracy vs. k (Test Set, uniform weights, Euclidean distance)

k	Accuracy	Precision	Recall	F1-Score
1	0.8990	0.8792	0.8623	0.8707
3	0.8979	0.8726	0.8678	0.8702
5	0.9077	0.8861	0.8788	0.8824
7	0.9088	0.8930	0.8733	0.8830
9	0.9088	0.8930	0.8733	0.8830
11	0.9099	0.9000	0.8678	0.8836

KDTree vs. BallTree ($k = 11$, uniform weights)

Metric	KDTree	BallTree
Accuracy	0.9099	0.9099
Training Time (s)	0.0167	0.0115
Prediction Time (s)	0.2659	0.2506

Tuned Model Performance (Test Set)

Metric	Value (Best KNN: $k = 9$, distance, manhattan, kd_tree)
Accuracy	0.9207
Precision	0.9421
Recall	0.8512
F1-score	0.8944
ROC-AUC	0.9712

5-Fold Cross-Validation Results (Training Set)

Fold	Bernoulli Naïve Bayes	Tuned KNN
1	0.9063	0.9389
2	0.8913	0.9280
3	0.8723	0.9253
4	0.8682	0.9266
5	0.8981	0.9076
Average	0.8872 $\pm$ 0.0147	0.9253 $\pm$ 0.0101

Theoretical Time Complexity

Algorithm	Training	Prediction
Naïve Bayes	$O(nd)$	$O(d)$
KNN (Brute)	$O(1)$	$O(nd)$
KDTree	$O(n \log n)$	$O(\log n)_{avg}$
BallTree	$O(n \log n)$	$O(\log n)_{avg}$

Here $n = 3680$ (training samples) and $d = 57$

(features).

Experimental Time Analysis

Algorithm	Training Time (s)	Prediction Time (s, full test set)
Gaussian NB	0.00589	0.00194
Multinomial NB	0.00277	0.00043
Bernoulli NB	0.00527	0.00143
Tuned KNN (kd_tree)	0.01600	0.28618

Additional Task: Distance Metric Comparison ($k = 9$, uniform weights)

Metric	Accuracy	Precision	Recall	F1-Score
Euclidean	0.9088	0.8930	0.8733	0.8830
Manhattan	0.9034	0.9335	0.8127	0.8689

Additional Task: Weighted KNN Comparison ($k = 9$)

Weights	Accuracy	Precision	Recall	F1-Score
Uniform	0.9088	0.8930	0.8733	0.8830
Distance	0.9164	0.8972	0.8898	0.8935

Additional Task: Train-Test Split Sensitivity

Split	Test Accuracy (Bernoulli NB)	Test Accuracy (Tuned KNN)
70/30	0.8870	0.9312
80/20	0.8762	0.9207
90/10	0.8785	0.9067

10 Result Visualization

Confusion Matrices on Test Set

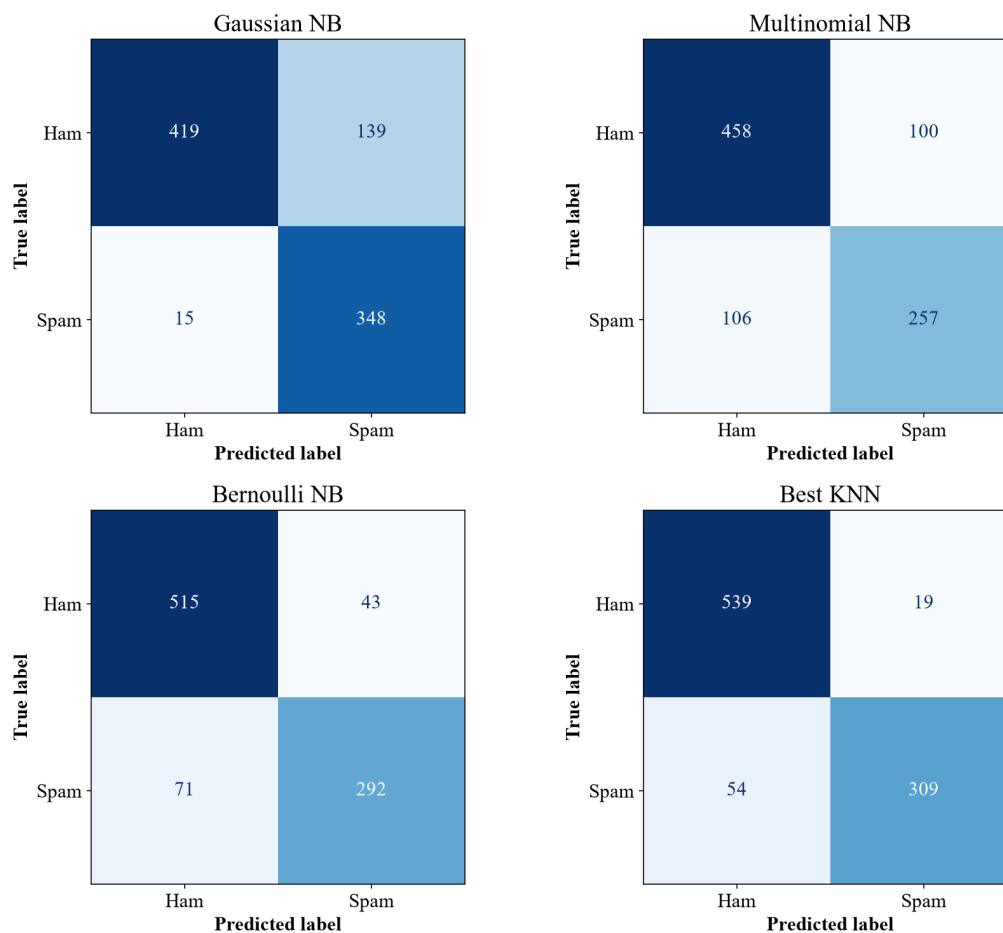


Figure 2: Confusion matrices for all four classifiers on the test set.

Inference: Gaussian NB has the fewest false negatives (missed spam) but the most false positives (ham wrongly marked spam), consistent with its high recall / low precision profile. Bernoulli NB and the tuned KNN show the most balanced confusion matrices, with KNN achieving the fewest total misclassifications (73 of 921 test samples).

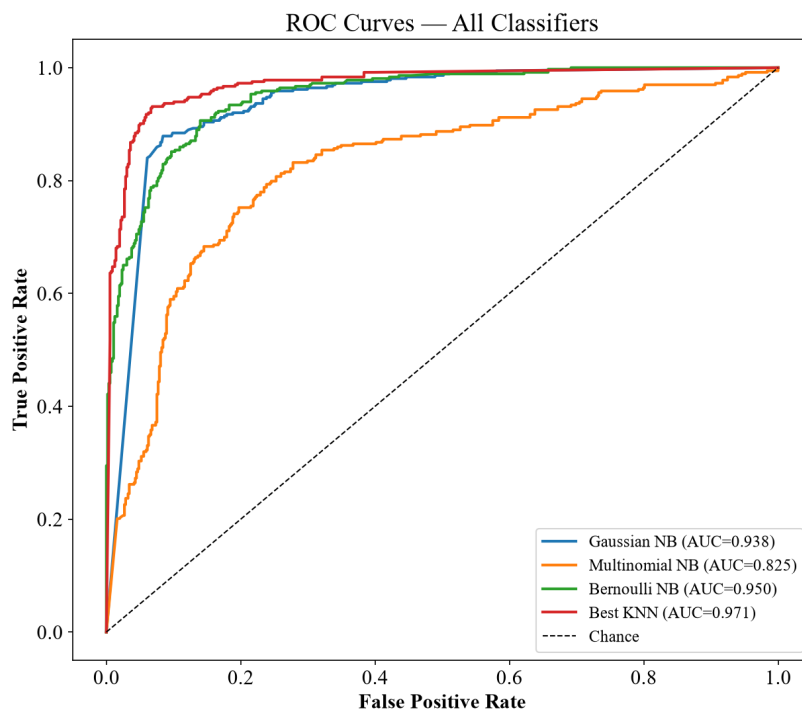


Figure 3: ROC curves for all classifiers.

Inference: The tuned KNN model achieves the highest AUC (0.971), followed closely by Bernoulli NB (0.950) and Gaussian NB (0.938); Multinomial NB trails with an AUC of 0.825. All curves lie well above the diagonal, confirming that every classifier has genuine discriminative power over random guessing.

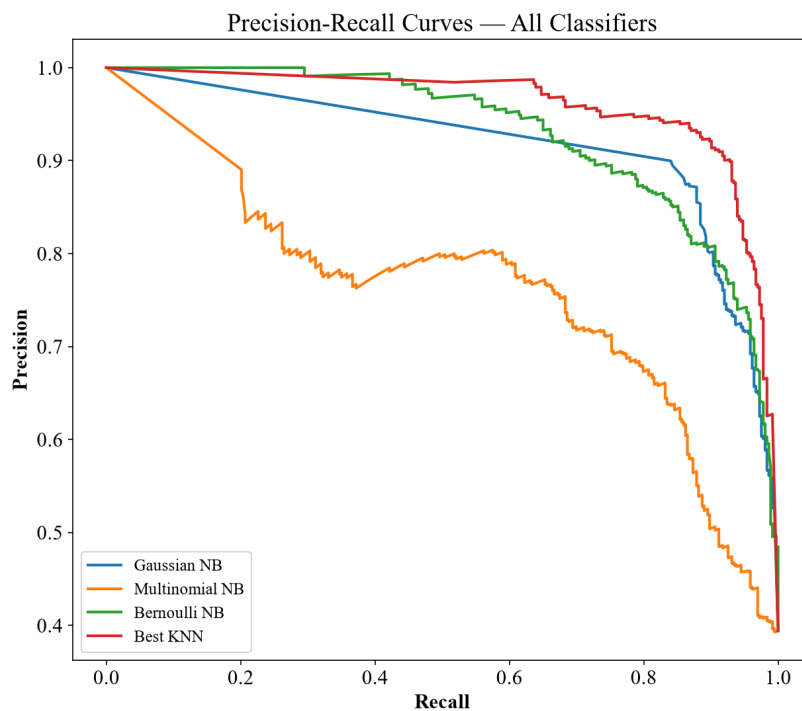


Figure 4: Precision-Recall curves for all classifiers.

Inference: Because spam is the minority class, the PR curve is a stricter test than ROC. KNN and Bernoulli NB maintain high precision across most of the recall range, while Gaussian NB's precision drops sharply as recall approaches 1; it achieves high recall only by aggressively over-predicting the spam class.

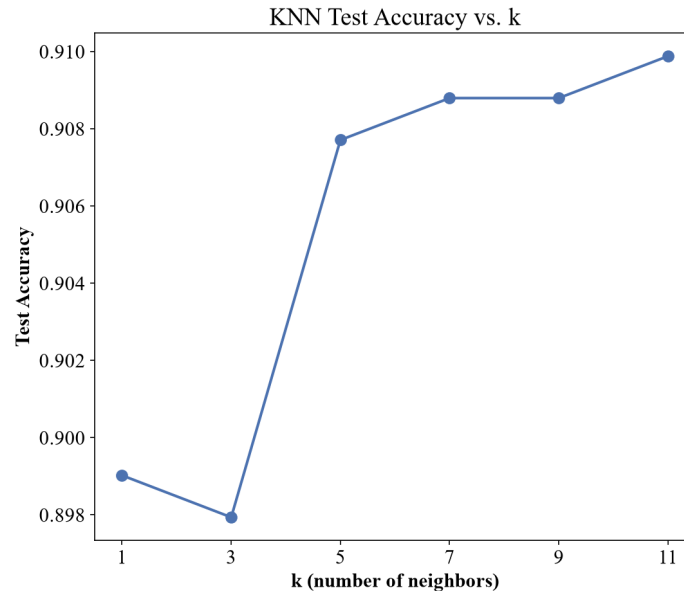


Figure 5: KNN test accuracy as a function of k .

Inference: Accuracy rises from $k = 1$ to around $k = 7$ – 11 and then plateaus, indicating that very small k values (especially $k = 1$) are noisier / more prone to overfitting individual training points, while accuracy stabilizes once enough neighbors are averaged to smooth out noise, without yet showing underfitting.

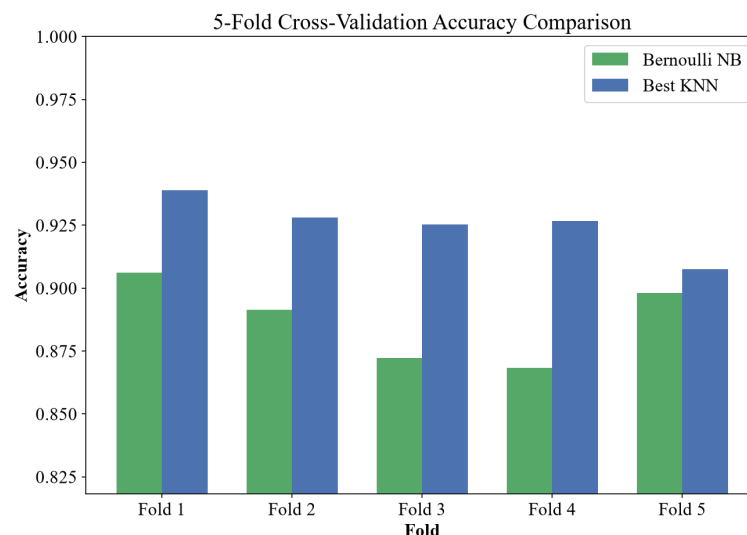


Figure 6: 5-fold cross-validation accuracy: Bernoulli NB vs. tuned KNN.

Inference: The tuned KNN outperforms Bernoulli NB in every single fold, by a consistent margin of roughly 3–5 percentage points, so its advantage on the held-out test

set is not a one-off artifact of a single split.

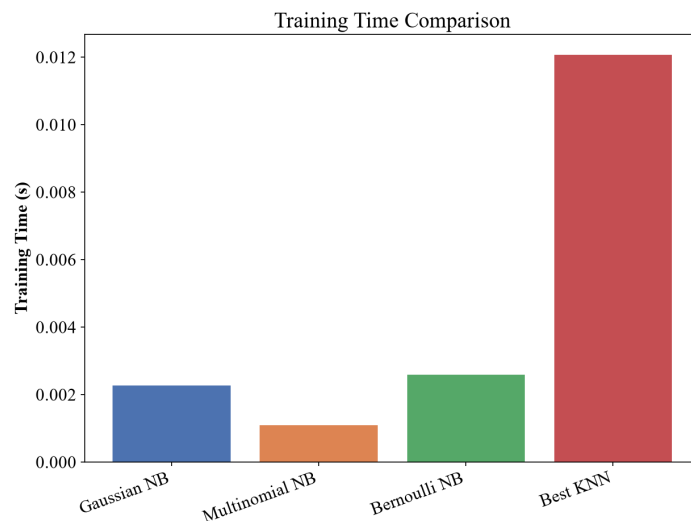


Figure 7: Training time comparison across all four classifiers.

Inference: All Naïve Bayes variants train in under 6 milliseconds since they only compute closed-form summary statistics. KNN’s “training” time (here, building the KD-tree index) is also small (16 ms) since it does no real learning, confirming the $O(1)/O(n \log n)$ theoretical expectation.

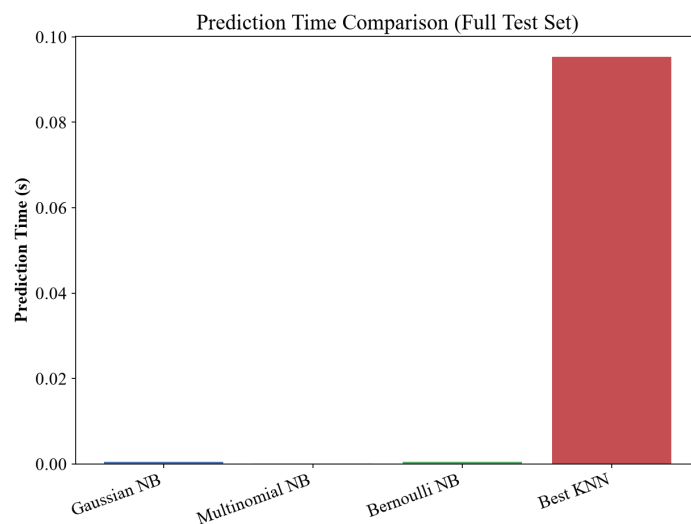


Figure 8: Prediction time comparison (full 921-sample test set).

Inference: KNN prediction is roughly 150–600× slower than any Naive Bayes variant, since every query requires a nearest-neighbor search across the training set. This is the clearest practical illustration of the $O(nd)/O(\log n)_{avg}$ vs. $O(d)$ complexity gap noted in the theoretical table.

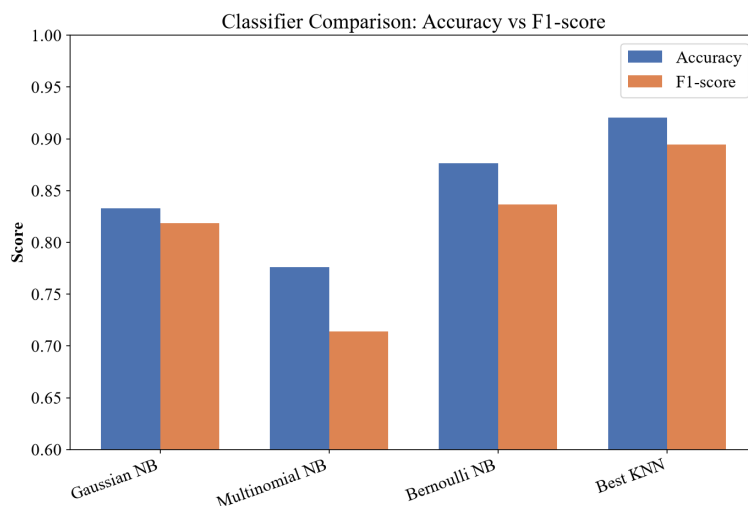


Figure 9: Overall Accuracy and F1-score comparison across all classifiers.

Inference: The tuned KNN is the best all-round performer (highest accuracy and F1), Bernoulli NB is a close, much faster runner-up, and Multinomial NB is the weakest model on this particular feature representation.

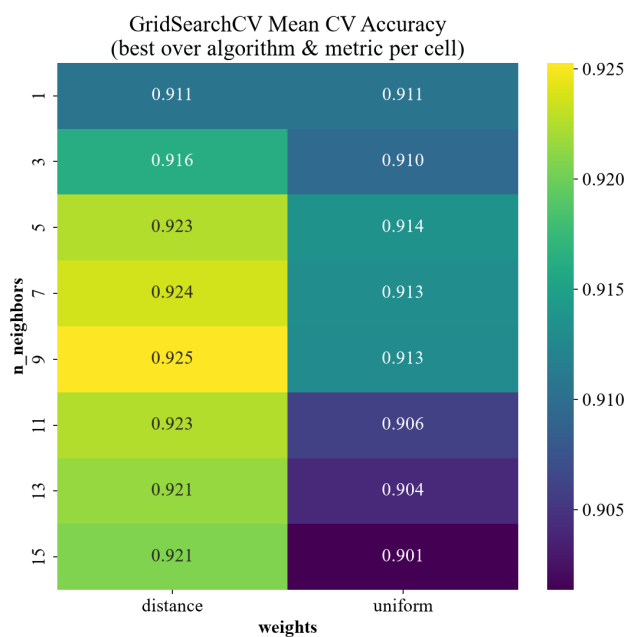


Figure 10: GridSearchCV mean cross-validation accuracy across $n_neighbors$ and $weights$ (best value over algorithm/metric shown per cell).

Inference: Distance-weighted KNN consistently outperforms uniform weighting at every k , and accuracy peaks in the $k = 7$ – 11 range before flattening, guiding the search toward the reported optimum of $k = 9$.

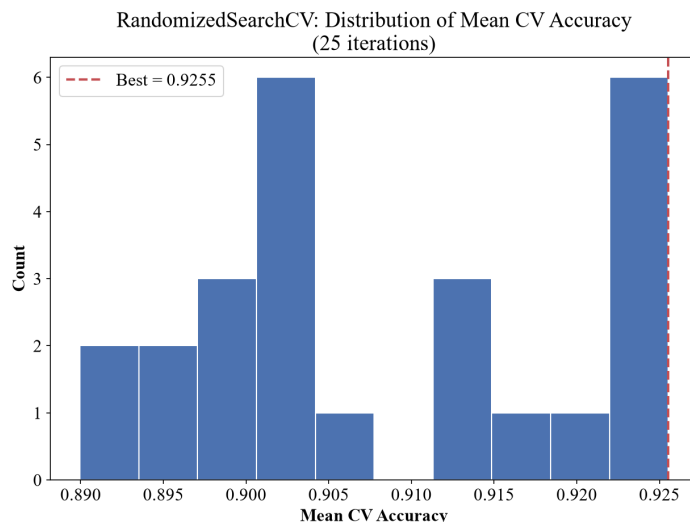


Figure 11: Distribution of mean CV accuracy across the 25 RandomizedSearchCV iterations.

Inference: Most sampled configurations already achieve accuracy above 0.90, and the best sampled score (0.9255) is statistically indistinguishable from the exhaustive grid search optimum (0.9253), showing that random sampling of roughly 20% of the search space was sufficient to find a near-optimal configuration in less than half the compute time.

11 Discussion

Analysis Questions

1. **Which Naïve Bayes variant performed best?** Bernoulli NB, with the highest accuracy (0.876), F1-score (0.837) and ROC-AUC (0.950) among the three variants. Its binarize-then-model-presence/absence approach fits the heavily zero-inflated, right-skewed nature of the Spambase features better than Gaussian NB’s normality assumption or Multinomial NB’s pure count model.

2. **What is the optimal value of k ?** Both search strategies converge to $k \approx 8$ –9 with distance-weighting and the Manhattan metric, achieving a 5-fold CV accuracy of about 0.925, noticeably higher than the 0.910 obtained with the default uniform-weight, Euclidean-distance configuration at the same k .

3. **Compare GridSearchCV and RandomizedSearchCV.** GridSearchCV exhaustively evaluated all $8 \times 2 \times 2 \times 2 = 128$ combinations in 58.5 s and found a CV accuracy of 0.9253. RandomizedSearchCV sampled only 25 combinations in 23.8 s (59% faster) and found a marginally higher score of 0.9255. For this search space the two methods are effectively equivalent in quality, but RandomizedSearchCV is clearly more time-efficient.

4. **Compare KDTree and BallTree.** Both produce identical predictions and accuracy (they are exact, not approximate, neighbor searches), differing only in construction/query speed. On this dataset BallTree built marginally faster (0.0115 s vs. 0.0167 s) and queried marginally faster (0.251 s vs. 0.266 s), though the difference is small enough to be within run-to-run noise.

5. Compare theoretical and practical complexity. The measured times track the theoretical expectations well: Naïve Bayes trains and predicts in single-digit milliseconds regardless of variant ($O(nd)/O(d)$), while KNN prediction (0.25–0.29 s) is orders of magnitude slower than any NB variant, consistent with its $O(nd)$ brute-force / $O(\log n)_{avg}$ tree-based query cost. However, the tree-based speed-up over brute force is muted here because $d = 57$ is high enough that KDTree/BallTree partitioning becomes less effective (the curse of dimensionality); both trees still required a meaningful search rather than a near- $O(\log n)$ lookup.

6. Which classifier is preferred for large datasets? Naïve Bayes, because its training and prediction costs scale linearly in n and d with very small constants and no need to retain the full training set at inference time, whereas KNN’s prediction-time cost grows directly with the number of stored training points, becoming a bottleneck as n grows.

Additional Tasks

- **Train-test split sensitivity:** Accuracy for both models was stable across 70/30, 80/20 and 90/10 splits (Bernoulli NB: 0.877–0.887; tuned KNN: 0.907–0.931), indicating the results are not an artifact of one particular split. Both models perform best with more training data relative to test data (70/30), as expected.
- **Euclidean vs. Manhattan distance:** Manhattan distance gave higher precision but lower recall than Euclidean at $k = 9$; Euclidean gave the better overall F1-score (0.883 vs 0.869) in the fixed-weights comparison, though the grid search found that when combined with *distance* weighting, Manhattan actually produced the best overall configuration, showing that hyperparameters interact rather than acting independently.
- **Weighted KNN:** Distance-weighting improved accuracy, recall and F1-score over uniform weighting at the same k , since it lets nearby neighbors dominate the vote and reduces the influence of ties among more distant points.
- **Overfitting/underfitting vs. k :** $k = 1$ effectively memorizes the training data (zero bias, high variance) and is the most overfit-prone setting, visible here as the lowest test accuracy in the sweep. As k increases to 7–11, variance drops and accuracy improves; had k been pushed much higher (e.g., $k \rightarrow n$), the decision boundary would flatten toward predicting the majority class everywhere, causing underfitting (rising bias). The plateau observed between $k = 7$ and $k = 11$ suggests this dataset’s sweet spot lies in that range.

Bias-Variance Trade-off

Naïve Bayes classifiers are high-bias, low-variance models: their strong independence assumption limits how well they can fit complex decision boundaries (visible in Multinomial NB’s comparatively low accuracy), but this same rigidity makes them stable across resampling, as reflected in Bernoulli NB’s tight cross-validation spread (std = 0.0147). KNN is a low-bias, higher-variance model whose flexibility (especially at small k) lets it fit the training distribution closely, which is why tuning k and the weighting scheme had a measurable effect on its performance and why it ultimately out-performed every Naïve Bayes variant once properly tuned.

12 Conclusion

Both families of classifiers achieved strong performance on the Spambase dataset, but with different trade-offs. Bernoulli Naïve Bayes was the best-performing and by far the fastest of the Naive Bayes variants, thanks to its binarized, presence/absence modelling of the skewed, zero-inflated feature set. The tuned KNN classifier ($k = 9$, distance-weighted, Manhattan distance) achieved the best overall accuracy (0.921) and ROC-AUC (0.971) of all four models, at the cost of roughly two orders of magnitude higher prediction latency. RandomizedSearchCV located an equally good configuration to GridSearchCV in under half the time, making it the more practical choice for larger hyperparameter spaces. For a production spam filter prioritizing throughput, Bernoulli NB offers the best accuracy-per-millisecond; for a setting where prediction latency is less critical than raw accuracy, the tuned KNN model is preferable.

13 References

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